c) four equivalent orbitals, that are lying the same plane will be formed

d) none of these

11. Which of these represents the correct order of their increasing bond order.

a) $C_2 < C_2^{2-} < O_2^{2-} < O_2$

b) $C_2^{2-} < C_2^{+} < O_2 < O_2^{2-}$

c) $O_2^{2-} < O_2 < C_2^{2-} < C_2^{+}$

d) $O_2^{2-} < C_2^{+} < O_2 < C_2^{2-}$

12. Hybridisation of central atom in PCl_5 involves the mixing of orbitals.

a) s, p_x, p_y, d_{x2}, d_{x2-y2}

b) s, p_x. p_y, p_{xy}. d_{x2-y2}

c) s, p_x, p_y, p_z, d_{x2-y2}

d) s, p_x, p_y, d_{xy}, d_{x2-y2}

13. The correct order of O-O bond length in hydrogen peroxide, ozone and oxygen is

a) $H_2O_2 > O_3 > O_2$

b) $O_2 > O_3 > H_2O_2$

c) $O_2 > H_2O_2 > O_3$

d) $O_3 > O_2 > H_2O_2$

14. Which one of the following is diamagnetic.?

a) O_2 b) O_2^{2-}

c) O_2^{+} d) None of these

15. Bond order of a species is 2.5 and the number of electons in its bonding molecular orbital is formd to be 8 The no. of electons in its antibonding molecular orbital is

a) three b) four

c) Zero d) can not be calculated from the given information.

16. Shape and hybridisation of IF_5 are

a) Trigonal bipyramidal, Sp^3d^2

b) Trigonal bipyramidal, Sp^3d

c) Square pyramidal, Sp^3d^2

d) Octahedral, Sp^3d^2

17. Pick out the incorrect statement from the following

a) Sp^3 hybrid orbitals are equivalent and are at an angle of 109^0 28' with eachother

b) dsp^2 hybrid orbitals are equivalent and bond angle between any two of them is 90^0

c) All five sp^3d hybrid orbitals are not equivalent out of these five sp^3d hybrid orbitals, three are at an angle of 120^0, remainir two are perpendicular to the plane containing the other three

d) none of these

18. The molecules having same hybridisation, shape and number of lone pairs of electons are

a) SeF_4, XeO_2F_2 b) SF_4, XeF_2

c) $XeOF_4$, TeF_4 d) $SeCl_4$, XeF_4

19. In which of the following molecules / ions BF_3, NO_2^-, H_2O the central atom is sp^2 hybridised?

a) NH_2^- and H_2O

b) NO_2- and H_2O

c) BF_3 and NO_2-

d) BF_3 and NH_2^-

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